IILM UNIVERSITY
GREATER NOIDA, UTTAR PRADESH

AICTE – EDUSKILLS VIRTUAL INTERNSHIP

AI Deployment and automation

An 8-Week Professional Internship Presentation Report

INTERNSHIP DURATION
June 2026 – August 2026 (8 Weeks)

STUDENT CREDENTIALS

Aman Sharma

Enrollment / Roll No.
2410030051

Program & Section
B.Tech (CSE) | 3CSE1

HOST ORGANIZATION
EduSkills Academy
Supported by AICTE National Internship Portal

Candidate & Program Overview

Candidate Details

Student Name: AMAN SHARMA

Roll Number: 2410030051

Degree: B.Tech (Computer Science & Engineering)

Academic Batch: 2026-2027 | Section: 3CSE1

University: IILM University, Greater Noida, U.P.

Internship Profile

Program Name: AICTE – EduSkills Virtual Internship

Specialization Domain: AI Deployment & Automation

Duration: 8 Weeks (June 2026 – August 2026)

Certificate ID: 2026-E31C2B5F43

Offer Letter Ref: C17Y2026M081522484

Organization Profile: EduSkills Academy

EduSkills Academy is an ISO 9001:2015 and ISO 20000-1:2018 certified organization operating under the vision of *"Nation Building Through Skills"*.

- **AICTE Collaboration:** Operates under the patronage of the AICTE National Internship Portal to empower engineering talent across India.
- **Industry Alignment:** Partners with leading global tech organizations to bridge academia and industry gaps in emerging tech.
- **Core Focus Domains:** Artificial Intelligence, MLOps, Cloud Computing, Cybersecurity, and Enterprise Automation.

Robotic Process Automation and AI Dashboard

Problem Statement & Industry Context

⚠ The Industry Challenge

While academic curriculum extensively covers building Machine Learning and Deep Learning models, students often lack hands-on exposure to deploying models into real-world production systems.

A high-performing ML model in a Jupiter Notebook provides zero business value until it is exposed as a reliable, scalable, monitorable, and secure API service.

🎯 The Solution Scope

This internship addresses the complete end-to-end lifecycle of production AI: model serving, container packaging, cloud hosting, automated scaling, observability, and retraining triggers.

Mastering industry-grade tools (FastAPI, Docker, Kubernetes, LangChain, SageMaker) to transform research prototypes into enterprise AI solutions.

Core Project Objectives

API & Containerization

Learn advanced model serving with FastAPI (async operations) and build portable, production-ready Docker containers for MLOps.

Cloud & Orchestration

Deploy AI/GenAI models on Amazon SageMaker, Vertex AI, and Azure ML, and manage multi-container pods using Kubernetes.

Agentic Workflows & RAG

Construct intelligent AI agents and business automation pipelines using LangChain, LlamaIndex, LangGraph, and Vector Databases.

Observability & Drift

Implement continuous monitoring and logging using Prometheus, Grafana, and Evidently AI for automated drift detection.

Technologies & MLOps Toolchain

Serving & Apps

Languages & APIs: Python, FastAPI (Async REST Services)

Front end Integration: Streamlit, Gradio Web UI

DevOps & Cloud

Containers: Docker, Kubernetes Cluster (Pods, Deployments)

Cloud Platforms: AWS SageMaker, Google Vertex AI, Azure ML

GenAI & Agents

Frameworks: LangChain, LlamaIndex, LangGraph

Serving & RAG: vLLM, Vector DBs, Quantization

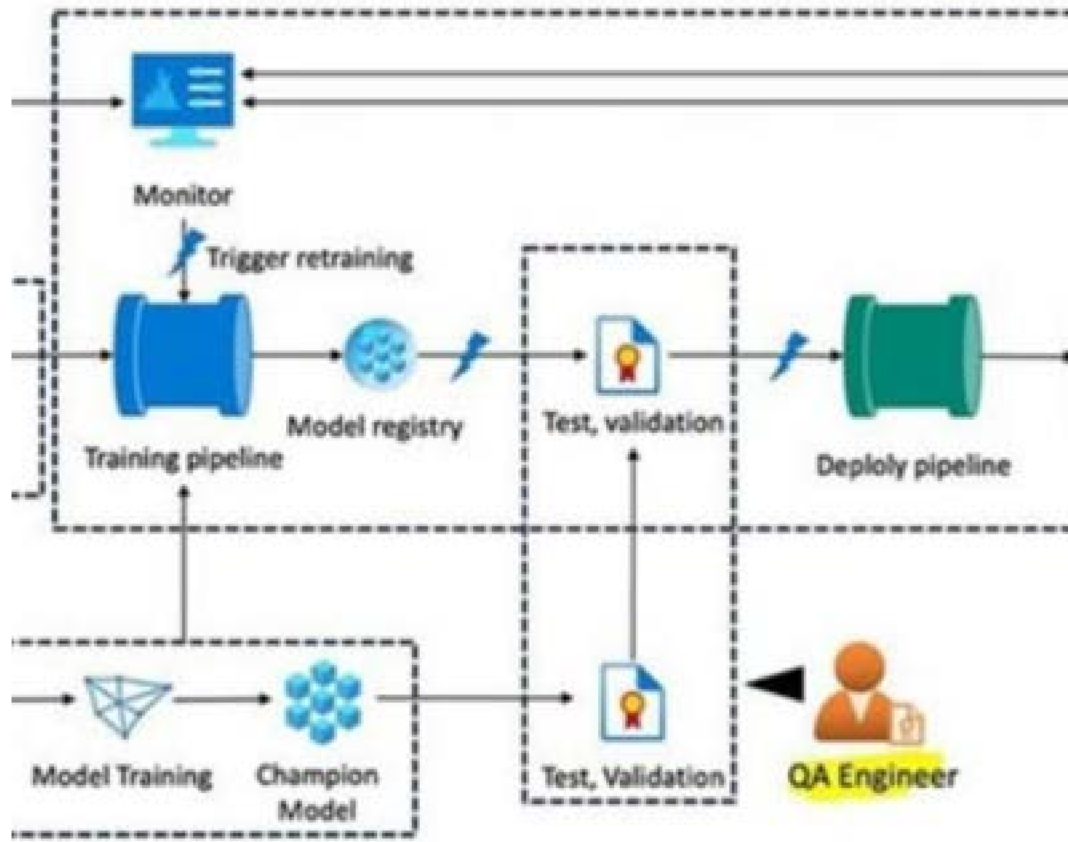
Monitoring & Drift

Observability: Prometheus, Grafana Dashboards

Drift Analysis: Evidently AI for data drift & retraining

Production AI Architecture

- **Model Layer:** Base ML / LLM models providing intelligence.
- **API Layer:** FastAPI wrapper for sync & async inference calls.
- **Container & Cloud:** Docker packaged containers orchestrated via Kubernetes on AWS/Azure/Vertex.
- **Agent & App Layer:** LangChain & LlamaIndex workflows integrated with Streamlit interfaces.
- **Observability Layer:** Real-time tracking with Prometheus, Grafana, and Evidently AI.



Curriculum Timeline (Weeks 1 to 4)

Week 1

AI Deployment & FastAPI

AI lifecycle, MLOps terminology, async API endpoints, and production readiness.

Week 2

Docker & CI/CD

Dockerfiles, container volumes, networking, and automated CI/CD for machine learning.

Week 3

Cloud AI & GenAI

SageMaker, Vertex AI, Azure ML, vLLM serving, and model quantization techniques.

Week 4

AI Agents & Automation

LangChain, LlamaIndex, agent orchestration using LangGraph workflows.

Curriculum Timeline (Weeks 5 to 8)

Week 5

Security & App Integration
Streamlit/Gradio apps, model security, privacy, and bias mitigation strategies.

Week 6

Kubernetes & Observability
Container orchestration, Pods, Services, Prometheus, and Grafana logging.

Week 7

Optimization & Drift
Evidently AI drift detection, auto-retraining, Advanced RAG, and Vector DBs.

Week 8

Applied Capstone
End-to-end full-stack RAG app deployment, business automation AI agents.

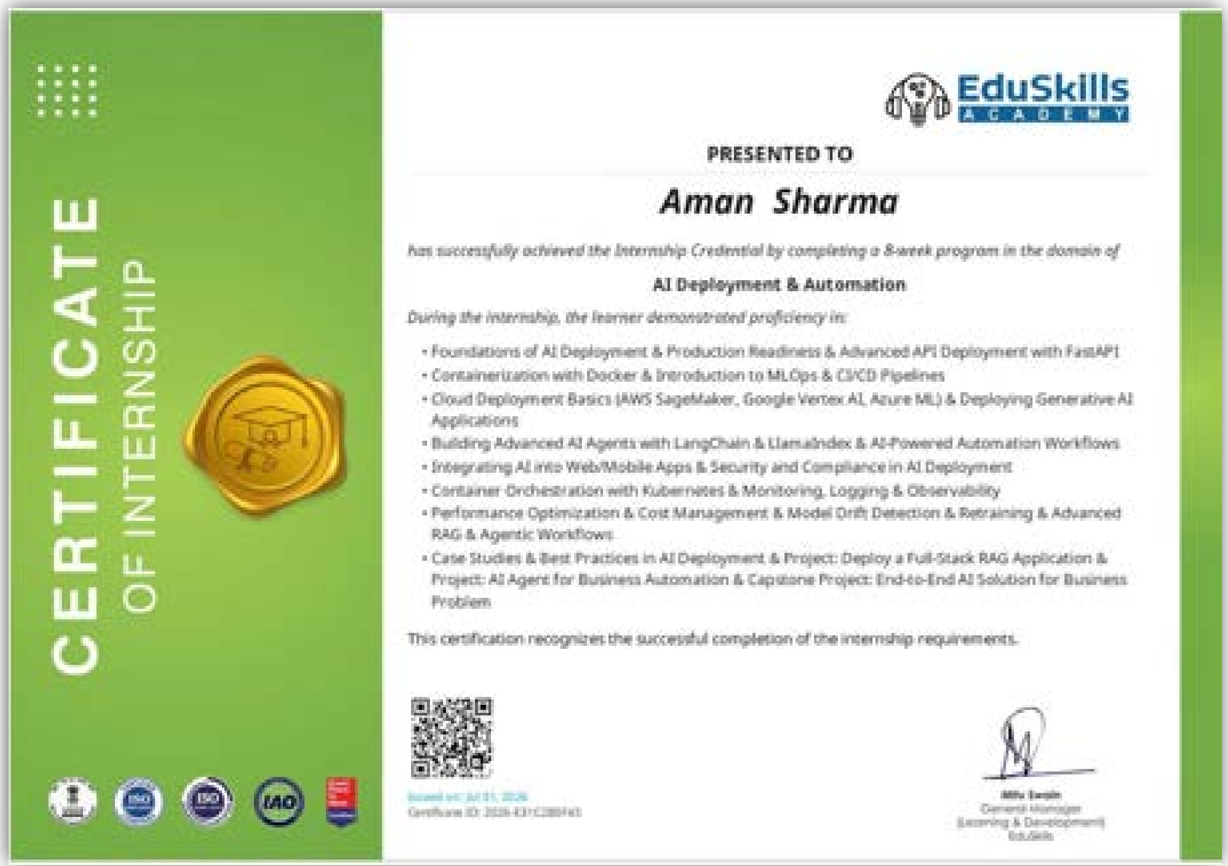
Key Projects & Demonstrated Skills

- **Production-Ready API Development:** Developed high-throughput RESTful endpoints using FastAPI with async support for AI inference.
- **Containerization & Orchestration:** Package ML models into lightweight Docker containers and orchestrating them via Kubernetes Pods & Services.
- **Full-Stack RAG Application:** Built an end-to-end Retrieval-Augmented Generation application utilizing Vector Databases, FastAPI backend, and Streamlit frontend.
- **Agentic Workflows for Automation:** Designed multi-step AI agents using LangChain and LangGraph for real-world enterprise business automation.
- **Model Governance & Drift Monitoring:** Configured Evidently AI alongside Prometheus and Grafana dashboards to track prediction drift and trigger retraining.

Certification & Official Verification

Verification Metric	Official Record / Credential Details
Internship Title	AI Deployment & Automation Virtual Internship
Issuing Organization	EduSkills Academy (ISO 9001:2015 & ISO 20000-1:2018 Certified)
Academic Patronage	AICTE National Internship Portal & EduSkills Foundation
Certificate ID	2026-E31C2B5F43 (Issued: Jul 31, 2026)
Offer Letter Ref	C17Y2026M081522484
AICTE Student ID	STU69e50fbaba77a1776619450

Certification & Official Verification



CERTIFICATE

OF INTERNSHIP

EduSkills

ACADEMY

PRESENTED TO

Aman Sharma

has successfully achieved the Internship Credential by completing a 8-week program in the domain of

AI Deployment & Automation

During the internship, the learner demonstrated proficiency in:

- Foundations of AI Deployment & Production Readiness & Advanced API Deployment with FastAPI
- Containerization with Docker & Introduction to MLOps & CI/CD Pipelines
- Cloud Deployment Basics (AWS SageMaker, Google Vertex AI, Azure ML) & Deploying Generative AI Applications
- Building Advanced AI Agents with LangChain & LlamaIndex & AI-Powered Automation Workflows
- Integrating AI into Web/Mobile Apps & Security and Compliance in AI Deployment
- Container Orchestration with Kubernetes & Monitoring, Logging & Observability
- Performance Optimization & Cost Management & Model Drift Detection & Retraining & Advanced RAG & Agentic Workflows
- Case Studies & Best Practices in AI Deployment & Project: Deploy a Full-Stack RAG Application & Project: AI Agent for Business Automation & Capstone Project: End-to-End AI Solution for Business Problem

This certification recognizes the successful completion of the internship requirements.

Thank You!

Aman Sharma
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School of Computer Science & Engineering
IILM University, Greater Noida, U.P.